

RescueGrid AI

Engineering Evaluation Document: AI-Powered Grid Resilience

AUDIT REPORT | Generated: 2026-07-02 21:27:33

1. Executive Summary

Grid Status:	ONLINE (Stable)	Active Disaster:	Normal (Sev: 0.00)
Total Renewables:	396.0 kW	Net Carbon Avoided:	16083.04 kg
Total Cost Savings:	■556942	Critical Infra Online:	100.0%

2. Engineering Validation Metrics

Metric	Value	Description
Renewable Penetration	40.7%	Percentage of total demand covered by renewable generation
Grid Reliability Index	100.0%	System Average Interruption equivalent for critical nodes
Battery State of Health	100.0%	Average effective capacity vs rated capacity (degradation tracked)
Avg Capacity Factor	0.1764	Average CF across all solar and wind assets
System Efficiency	245.9%	Ratio of useful consumption to total generation

3. Scenario Comparison: WITH vs WITHOUT AI

Comparing the RescueGrid AI optimized micro-grid performance against a standard diesel-backup baseline during the current scenario.

Metric	WITH RescueGrid AI	WITHOUT AI (Diesel)	Improvement
Hospital Avg Runtime	336.0 hrs	24.0 hrs	+1300.0%
Operating Cost	■ 66587.4	■ 1523620.8	-2188.2%
Carbon Emissions	823.07 kg	43109.2 kg	-5137.6%
Diesel Consumption	0.0 L	16085.5 L	100%
Critical Load Coverage	100.0%	100.0%	+0.0%

4. AI Optimization & Dispatch Log

The following logs detail the explainable AI decisions driven by the multi-strategy optimization engine.

Time	Action Type	Reasoning & Strategy Evaluation
15:57:23	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.7%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 187 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
15:57:23	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.7%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 187 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
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15:57:21	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.7%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 195 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
15:57:17	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.7%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 266 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
15:57:17	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.7%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 266 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
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15:57:12	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.7%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 391 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
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15:57:08	Optimization Completed	[OBSERVATION] Node SOC at 100.0%. Probabilistic risk of failure in 2h is 5.6%. [EVALUATION] Standard AHP Profile active (Cost: 50%, Rel: 30%, Sus: 20%). [ACTION] Force-route 463 kW of active renewables to this node's microgrid. Cost Score: 90.0 Reliability: 80.0 Sustainability: 100.0
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5. Sustainability Impact

Cost Analysis	Carbon & Environmental Equivalents
Generation Cost (Solar/Wind): ■ 66077.39 Diesel Baseline Cost: ■ 614904.62 Total Cost Avoided: ■ 548827.23	Equivalent Trees Planted: 765.9 trees Cars Off Road: 3.5 cars Households Powered: 25.17 homes/month